

**PLATES AND SHELLS UNDER PRESSURE
FORCES**

Part: **3.1 – STRENGTH OF METALLIC PARTS**

Chapter: **3190**

Title: **PLATES AND SHELLS UNDER
PRESSURE FORCES**

Revision: **0.3**

Date: **05/09/2023**

1 INTRODUCTION

1.1 METHOD-SUMMARY

1.2 SCOPE AND LIMITATIONS

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2 FLAT PLATES UNDER PRESSURE

Extracts of chapters 7.1 and 7.2 "PLATES" of Niu's book:

The plate shown in figure is a two-dimensional counterpart of the beam, in which transverse loads are resisted by flexural and shear stresses, with no direct stresses in its middle plane. The thin flat sheet, by deflecting enough to provide both the necessary curvature and stretch, may develop membrane stresses to support lateral loads. In the analysis of these situations no bending stress is presumed in the sheet.

Approximate criteria for classifying a plate subjected to normal loads may be classified in three groups based on relation of maximum deflection (δ) vs thickness (t):

- *Thick plate – supports the load by bending ($\delta/t < 0.5$)*
- *Membrane (or diaphragm) plate – supports the load by tension ($\delta/t > 5$)*
- *Thin plate – supports the load by both bending and tension and this plate is the primary class encountered in airframe structures ($0.5 < \delta/t < 5$)*

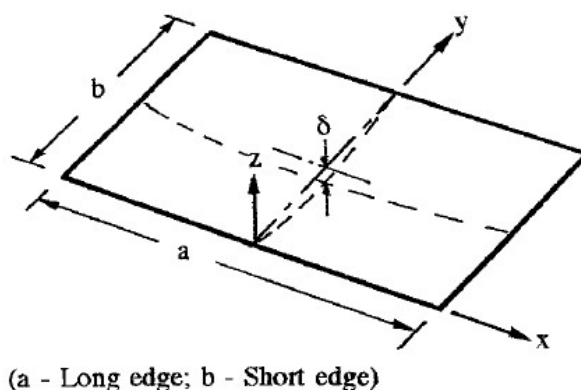


Figure 1. Plate subjected to normal loading (Ref. 2)

Since the membrane plate is seldom used on airframe applications it will not be discussed.

2.1 THICK PLATE

Technical literature offers graphical solutions for the maximum out-plane deflection and stresses of a rectangular plate subjected to uniform pressure, if this load is sustained in bending. The deflection calculated with this solution can be used to monitor if the thick plate hypothesis is valid or it is necessary to use more sophisticated analysis techniques. Expressions proposed by Niu (Ref. 2, §7.2(A)):

- Max. bending stress at centre: $\sigma_y = \beta \cdot p \cdot b^2 / t^2$ (*)
- Max. deflection at centre: $\delta = \alpha \cdot p \cdot b^4 / (E \cdot t^3)$
- Max. shear at centre of long plate a: $N_z = \kappa \cdot p \cdot b$

(*) Note the maximum bending stress in x direction shall not be higher than that value, i.e.: $\sigma_x \leq \sigma_y$

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Where coefficients α , β , κ can be obtained from the following graphs of Niu's book, Ref. 2 (which are equivalent to those in FIGURE 12 of ESDU 71013, Ref. 5)...

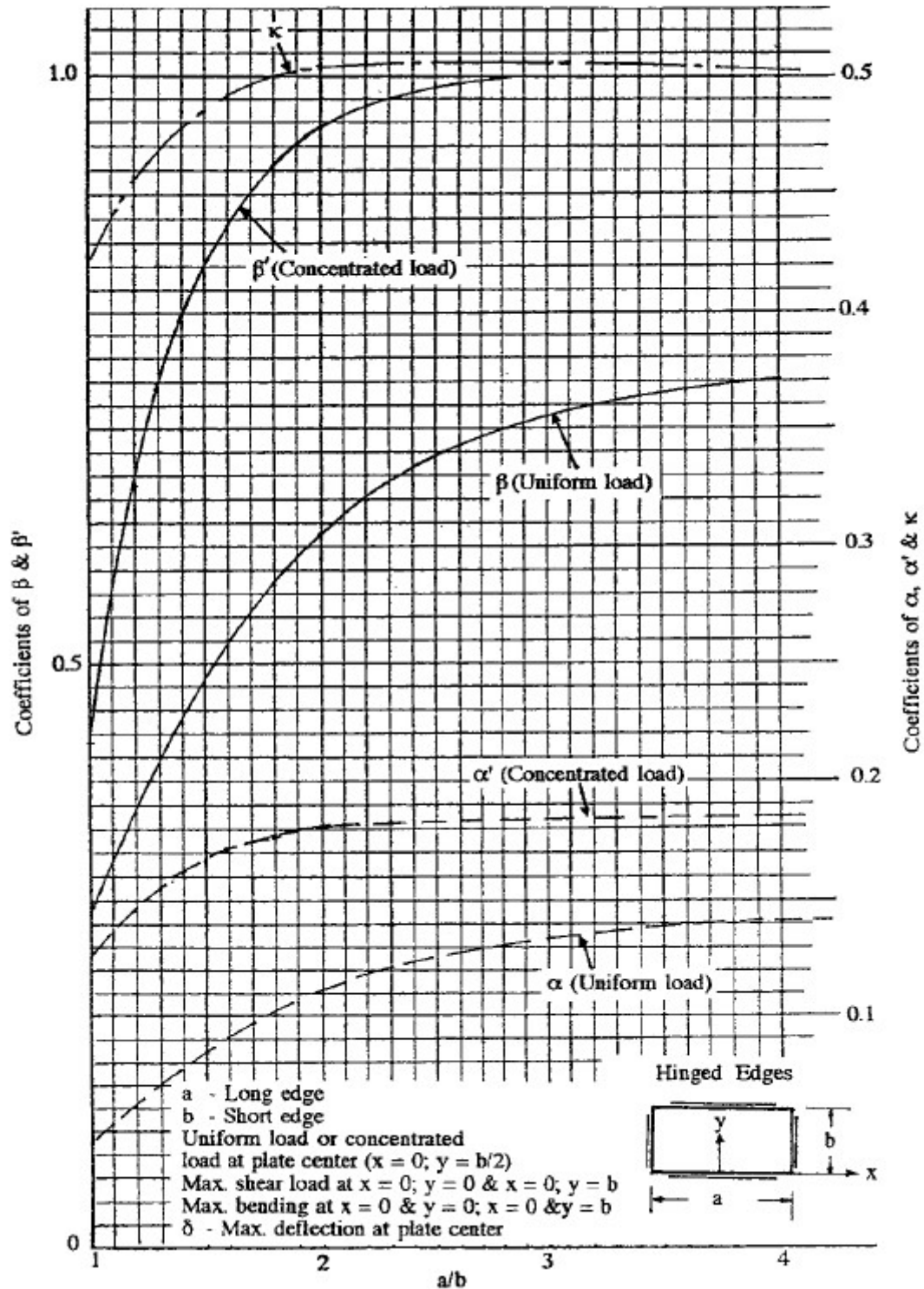


Figure 2. Coefficients α , β , κ for the simply supported thick plate (Niu's Fig. 7.2.2)

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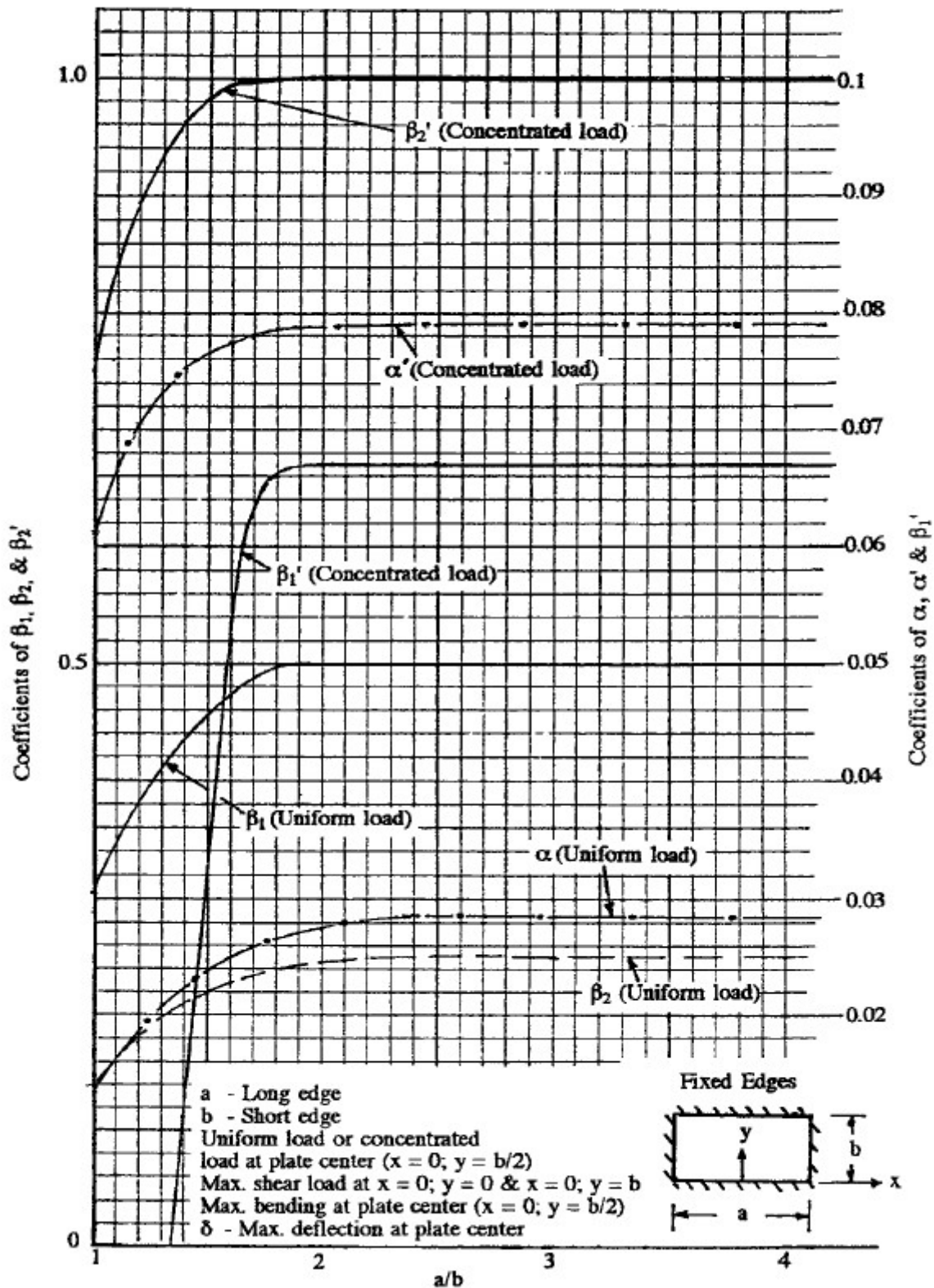


Figure 3. Coefficients α , β , κ for the clamped thick plate (Niu's Fig. 7.2.3)

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DEFENCE AND SPACE

Further notes

- ✓ For the thick plate, the superposition of stress states caused by pressure and by other sources (including in-plane loads) can typically be considered acceptable
- ✓ On the other hand, even the solutions discussed in previous pages do not take into account any non-linear effects due to material inelasticity, it is generally accepted as conservative to use the internal loads (plate forces and moments) provided by this theory at the critical locations as valid, and then calculate the stresses and static margins considering plastic bending.

2.2 THIN PLATE

Literature also offer some approximations of the solution for plates with large deflections, where pressure load is reacted partially in bending and partially in membrane/diaphragm stresses. ESDU 71013 (Ref. 5) or Roark's chapter 11 (Ref. 3) and examples of solutions dealing with that problem.

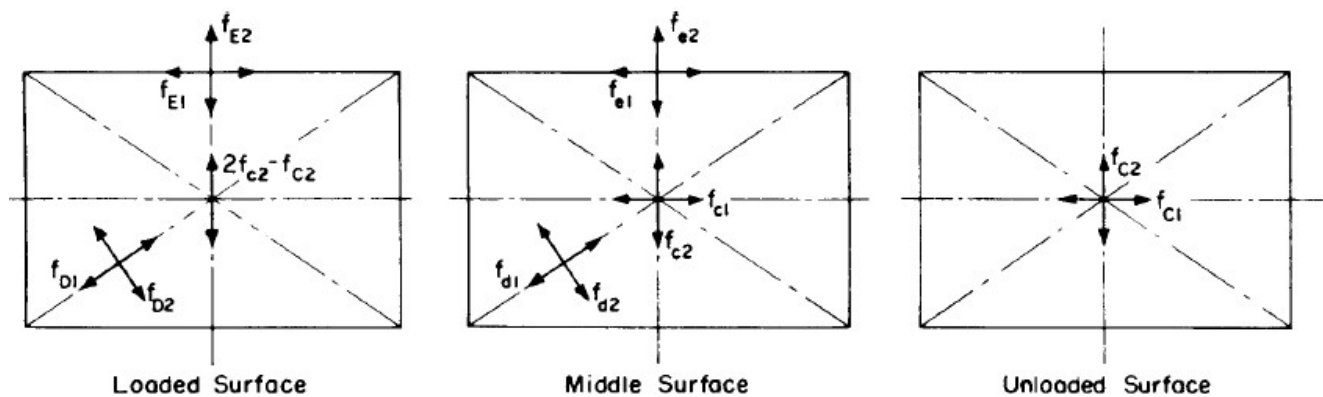


Figure 4. Points reported in ESDU 71013

Note that, for this case the problem is strongly non-linear and the superposition principle not applicable.

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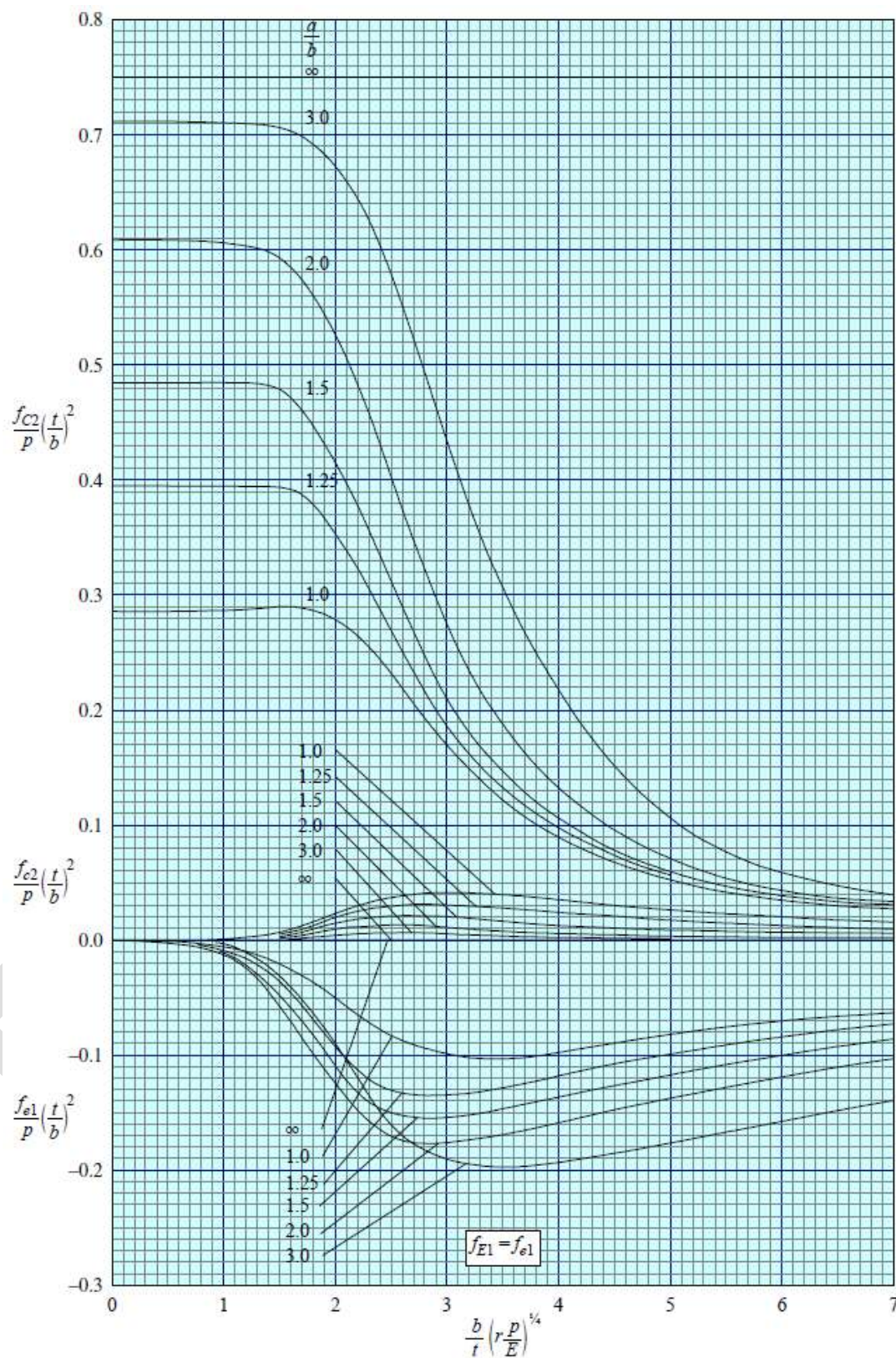


Figure 5. ESDU 71013 Figure 5: edges free in translation and rotation

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Rectangular plates under uniform load producing large deflection													
a/b	Edges and point of max σ	Coef.	qb^4/Et^4										
			0	12.5	25	50	75	100	125	150	175	200	250
1	Held, not fixed At center of plate $\left\{\begin{array}{l} \sigma_d b^2/Et^2 \\ \sigma b^2/Et^2 \end{array}\right\}$	γ/t	0	0.430	0.650	0.930	1.13	1.26	1.37	1.47	1.56	1.63	1.77
		0	0.70	1.60	3.00	4.00	5.00	6.10	7.00	7.95	8.60	10.20	
		0	3.80	5.80	8.70	10.90	12.80	14.30	15.60	17.00	18.20	20.50	
1	Held and riveted At center of plate $\left\{\begin{array}{l} \sigma_d b^2/Et^2 \\ \sigma b^2/Et^2 \end{array}\right\}$	γ/t	0	0.406	0.600	0.840	1.00	1.13	1.23	1.31	1.40	1.46	1.58
		0	0.609	1.380	2.68	3.80	4.78	5.75	6.54	7.55	8.10	9.53	
		0	3.19	5.18	7.77	9.72	11.34	12.80	14.10	15.40	16.40	18.40	
1	Held and fixed At center of long edges $\left\{\begin{array}{l} \sigma_d b^2/Et^2 \\ \sigma b^2/Et^2 \end{array}\right\}$	γ/t	0	0.165	0.32	0.59	0.80	0.95	1.08	1.19	1.28	1.38	1.54
		0	0.070	0.22	0.75	1.35	2.00	2.70	3.30	4.00	4.60	5.90	
		0	3.80	6.90	14.70	21.0	26.50	31.50	36.20	40.70	45.00	53.50	
	At center of plate $\left\{\begin{array}{l} \sigma_d b^2/Et^2 \\ \sigma b^2/Et^2 \end{array}\right\}$	0	0.075	0.30	0.95	1.65	2.40	3.10	3.80	4.50	5.20	6.50	
		0	1.80	3.50	6.60	9.20	11.60	13.0	14.50	15.80	17.10	19.40	
1.5	Held, not fixed At center of plate $\left\{\begin{array}{l} \sigma_d b^2/Et^2 \\ \sigma b^2/Et^2 \end{array}\right\}$	γ/t	0	0.625	0.879	1.18	1.37	1.53	1.68	1.77	1.88	1.96	2.12
		0	1.06	2.11	3.78	5.18	6.41	7.65	8.60	9.55	10.60	12.30	
		0	4.48	6.81	9.92	12.25	14.22	16.0	17.50	18.90	20.30	22.80	
2 to ∞	Held, not fixed At center of plate $\left\{\begin{array}{l} \sigma_d b^2/Et^2 \\ \sigma b^2/Et^2 \end{array}\right\}$	γ/t	0	0.696	0.946	1.24	1.44	1.60	1.72	1.84	1.94	2.03	2.20
		0	1.29	2.40	4.15	5.61	6.91	8.10	9.21	10.10	10.90	12.20	
		0	4.87	7.16	10.30	12.60	14.60	16.40	18.00	19.40	20.90	23.60	
1.5 to ∞	Held and fixed At center of long edges $\left\{\begin{array}{l} \sigma_d b^2/Et^2 \\ \sigma b^2/Et^2 \end{array}\right\}$	γ/t	0	0.28	0.51	0.825	1.07	1.24	1.40	1.50	1.63	1.72	1.86
		0	0.20	0.66	1.90	3.20	4.35	5.40	6.50	7.50	8.50	10.30	
		0	5.75	11.12	20.30	27.8	35.0	41.0	47.0	52.50	57.60	67.00	

Figure 6. Stresses of plates under pressure in large deflections (Roark's Ref. 3)

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2.3 COMBINED STRESSES

Pressure loads may be possibly combined with other types of forces, eg. in-plane loads. Direct superposition (addition) of stresses resulting from pressure cases with other, in particular with those coming from in-plane loads, is only permitted if the case is compliant with the “thick plate” condition. In other words, if the structure really behaves like a plate.

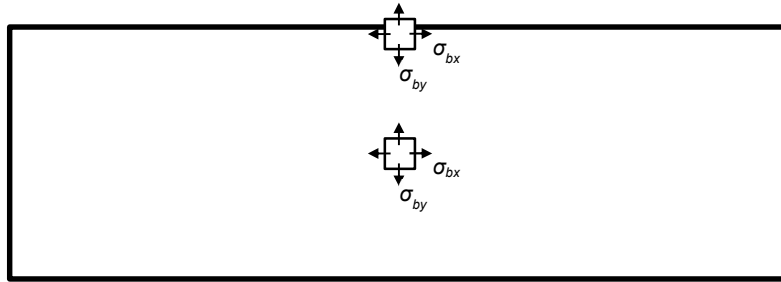


Figure 7. Critical location in plate under pressure: centre if hinged, centre of long edge if clamped

Then if behaviour is plate alike, the stress distribution coming from in-plane loads (uniform across the thickness) can be added to the bending stresses (varying across thickness) induced by pressure forces. Unless while material has not reached its yield stresses. For rectangular “thick plates” under pressure, handbook methodologies will normally provide only the most critical bending stresses. Using the subscript 0 to denote in-plane load/stress and b to denote pure bending:

- Simply supported (hinged) plate under uniform pressure

Bending stress shall be 0 at edges and critical at the centre of the plate. Assuming the X axis is that of the longest side of the plate, the most critical stress shall be σ_{by} , thus $|\sigma_{bx}| < |\sigma_{by}|$. The total stress at that point for the combined case, on the most outer faces:

$$\sigma_x = \sigma_{0x} \pm \sigma_{bx} \leq \sigma_{0x} \pm \sigma_{by}$$

$$\sigma_y = \sigma_{0y} \pm \sigma_{by}$$

$$\tau_{xy} = \tau_{0xy}$$

The familiar relationship between bending stress (σ_b), bending moment per unit width (m) and plate thickness (t) could be used, in preliminary sizing, to take into account approximately plastic bending for calculating the stress margins, if suitable tools are available:

$$\sigma_b = 6 \cdot m / t^2 \quad \rightarrow \quad m = \sigma_b \cdot t^2 / 6$$

Alternatively, Cozzone’s method (Bruhn C3.4) can be used to estimate the fictitious ultimate stress of the material in bending (rectangular section with $k = 1.5$, typ. $F_{ty} < f_m$):

$$F_b = f_m + f_0 (k - 1) \quad \rightarrow \quad F_{bu} = F_{tu} + 0.5 F_{ty} (k - 1)$$

The, if failure indices for in-plane and pressure loads are:

$$\checkmark \text{ in-plane : } R_0 = \sigma_{0vm} / F_{tu} = (\sigma_{0x}^2 - \sigma_{0x} \sigma_{0y} + \sigma_{0y}^2 + 3 \tau_{0xy}^2)^{0.5} / F_{tu}$$

$$\checkmark \text{ pressure : } R_b = \sigma_{0vb} / F_{bu} = (\sigma_{bx}^2 - \sigma_{bx} \sigma_{by} + \sigma_{by}^2)^{0.5} / F_{bu}$$

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(*) Through-thickness shear stresses due to distributed pressure shall normally not be significant and, in any case, zero at faces, where maximum bending occurs.

Finally, the Reserve Factor can be calculated assuming a linear interaction for in-plane and bending failures:

$$RF = 1 / (R_0 + R_b)$$

- Clamped plate under uniform pressure

In this case, the critical location shall be the centre of the long edges, the most critical stress shall be σ_{by} , and σ_{bx} shall be 0 at that point. The superposition of stresses get simpler than in previous case. The same technique for dealing for plastic bending (approximately) could be used in preliminary sizing.

2.4 HIGH FIDELITY STRESS METHODS

2.4.1 Energy Rayleigh-Ritz methods for the thick plate

Numerical solutions based in the energy Rayleigh-Ritz methodology, applied also to solve buckling stability problems, can be used also to treat the case of the thick plate under pressure (see Ref. 6). Implementations accounting for large displacements (diaphragm stresses) or material plasticity are, however, not currently available.

2.4.2 Finite Element Models

Finite Element Models shall generally capable to handle the most complicated problems of this kind. Details about modelization techniques, tools, etc... are out of the scope of this chapter.

3 FLAT STIFFENED PANELS

3.1 PRELIMINARY SIZING

The simplified approach to treat by analysis a flat stiffened panel under pressure load shall be to integrate the pressure across the stiffened section (considering the area of half adjacent sub-panels) and superimpose that stress status to that due to the in-plane forces). Figure below exemplify this strategy for a bulkhead frame supported on its attachments to fuselage skin.

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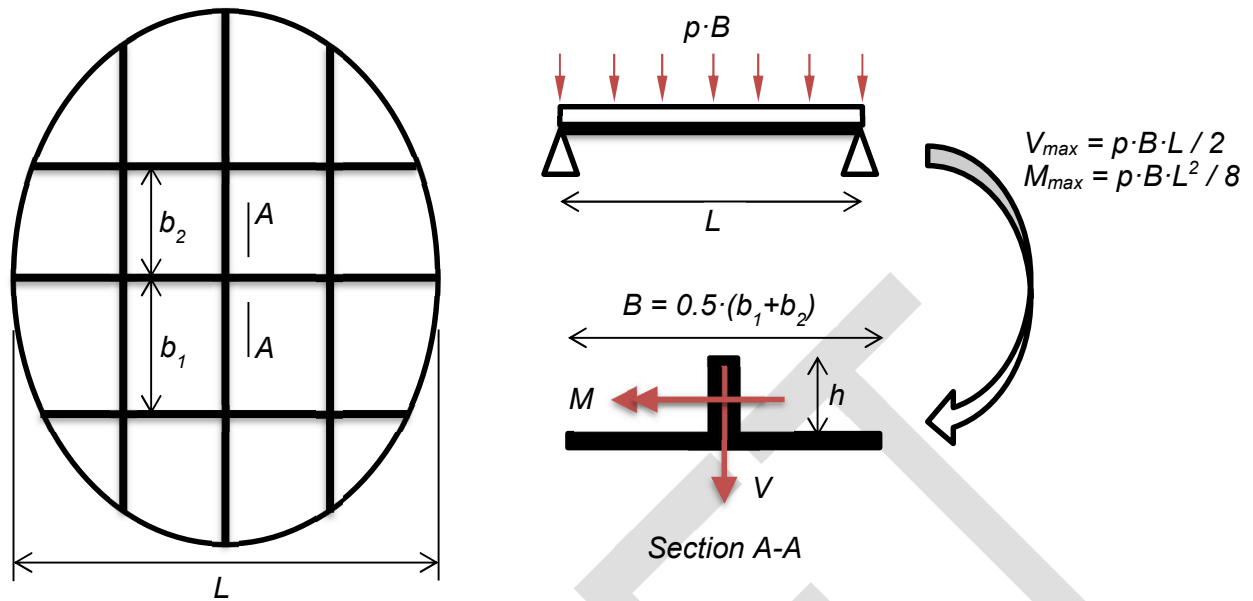


Figure 8. Integration of pressure forces in a flat stiffened panel

A strip of this structure, including a stiffener and the halves of the adjacent subpanels, can be treated as simply supported beam under the distributed load caused by the pressure forces. Critical internal loads on a representative section of this beam are shown above. Stress and strength of this section can be estimated by known methodologies.

Additionally, the subpanels delimited by the stiffeners should be analysed using the methods described in section §2.

3.2 HIGH FIDELITY STRESS METHODS

See discussion in sub-sections §2.4.1 and §2.4.2.

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4 CURVED SHELLS UNDER PRESSURE

Extract of chapter 7.3 "CYLINDRICAL SHELLS" of Niu's book: *Structurally, the most efficient form of pressure vessel is one in which the lateral pressures are supported by tensile stresses alone in the curved skins of the body, as shown in Fig. 7.3.1. The skins of these bodies have zero bending stiffness and hence have the properties of a membrane or diaphragm stress. The stresses develop, lying wholly in tangential direction at each point, are called membrane stresses.*

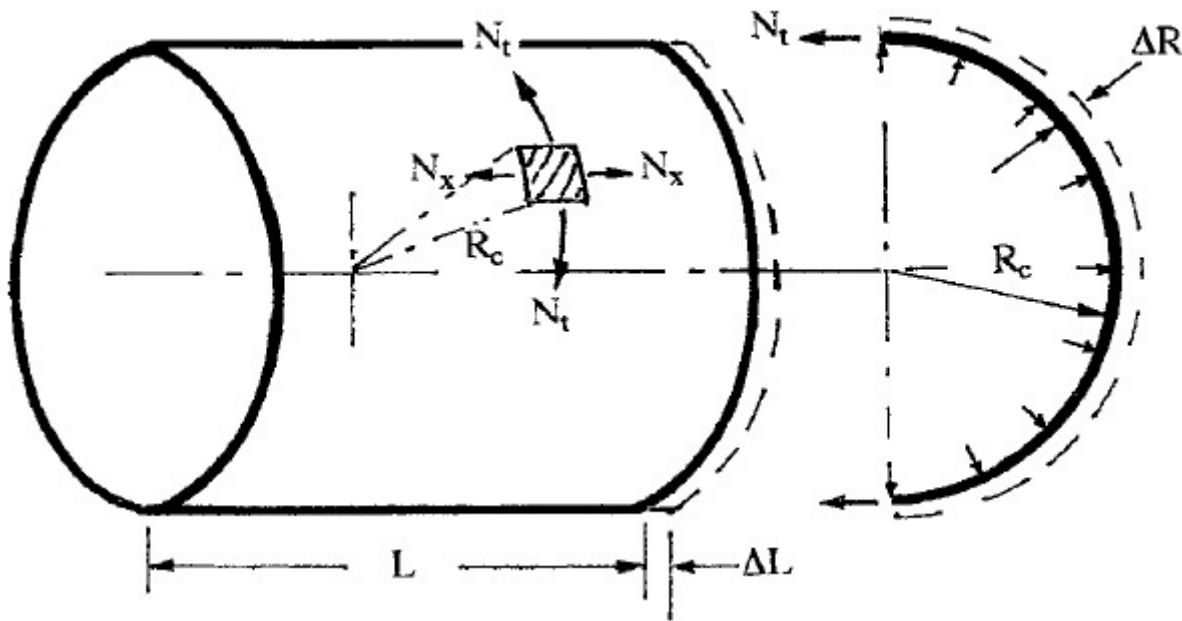


Figure 9. Cylindrical Shell (Ref. 2)

Basic equations for the internal load in a cylindrical shell (single curvature) subjected to pressure:

- The hoop tension load: $N_t = p \cdot R_c$
- The hoop tension stress: $\sigma_t = p \cdot R_c / t$

The above mentioned reference presents also the solution for the longitudinal X load assuming the shell belongs to a closed vessel. It is not shown here. For a shell belonging to a typical airframe (not a vessel), normally the greatest part of the X load shall be caused by global aerodynamic and inertial loads (e.g. global bending in a section of a fuselage).

4.1 CURVED STIFFENED PANELS AND PILLOWING EFFECT

Mirar lo de Airbus Comercial...

PLATES AND SHELLS UNDER PRESSURE FORCES

5 APPENDICES

ABBREVIATIONS AND SYMBOLS

<i>Symbol</i>	<i>Units</i>	<i>Definition</i>
a	mm	Longitudinal (X) dimension of a plate
b	mm	Transverse (Y) dimension of a plate
t	mm	Thickness of plate/shell
E	MPa	Young modulus of the material
p	MPa	Pressure difference on the plate or shell, For shells, sign convention is positive if inner value (concave side of shell) greater than outer value (convex side of shell)
δ	mm	Out-plate deflection of the plate
σ_x, σ_y	MPa	Stress in X and Y direction (typ. at a face of the plate)
R_c	mm	Curvature radius of the shell (assumed thin shell)
N_t	N/mm	Tangential/hoop load
σ_t	MPa	Tangential/hoop stress
TBC		To Be Confirmed

REFERENCES

	<i>Document Reference</i>	<i>Issue</i>	<i>Title</i>
Ref. 1	[E. F. Bruhn]	1973	ANALYSIS AND DESIGN OF FLIGHT VEHICLE STRUCTURES
Ref. 2	[Michael C. Y. Niu]	2 nd Ed.	AIRFRAME STRESS ANALYSIS AND SIZING
Ref. 3	[W. C. Young & R. G. Budynas]	7th Ed.	Roark's Formulas for Stress and Strain
Ref. 4	HSB 32412-02	Is. C	Normal stresses and deformations of flat rectangular plates loaded by a linearly distributed pressure
Ref. 5	ESDU 71013	1995	Elastic direct stresses and deflections for flat rectangular plates under uniformly distributed normal pressure
Ref. 6	NT-T-ADP-06002	Ed. A	Manual teórico del programa PANPA, versión 5.0

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SOFTWARE

<i>Program</i>	<i>Version</i>	<i>Description</i>

Table 1. Software programs referenced in this chapter

RECORD OF REVISIONS

<i>Revision Date</i>	<i>Authors</i>	<i>Change Reason</i>	<i>Pages/Sub-chapters affected</i>
0.3 05/09/2023	G. Baños	First issue	All pages

Table 2. Record of Revisions: authors, change reasons and sections/pages affected

APPROVAL SIGNATURES TABLE

<i>Dpt. \ Site</i>	<i>Getafe</i>	<i>Manching</i>
<i>Stress Methods (TEPAP-TL4)</i>	G. Baños DD/09/2023	-
<i>Structural Analysis (TEPAA)</i>	-	F. Glock DD/09/2023

Table 3. Approval signatures table for last revision of this chapter